MHR: A Multi-Modal Hyperbolic Representation Framework for Fake News Detection

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***Abstract*—The rapid growth of the internet has led to an alarming increase in the dissemination of fake news, which has had many negative effects on society. Various methods have been proposed for detecting fake news. However, these approaches suffer from several limitations. First, most existing works only consider news as separate entities and do not consider the correlations between fake news and real news. Moreover, these works are usually conducted in the Euclidean space, which is unable to capture complex relationships between news, in particular the hierarchical relationships. To tackle these issues, we introduce a novel Multi-modal Hyperbolic Representation framework (MHR) for fake news detection. Specifically, we capture the correlations between news for graph construction to arrange and analyze different news. To fully utilize the multi-modal characteristics, we first extract the textual and visual information, and then design a Lorentzian multi-modal fusion module to fuse them as the node information in the graph. By utilizing the fully hyperbolic graph neural networks, we learn the graph's representation in hyperbolic space, followed by a detector for detecting fake news. The experimental results on three real-world datasets demonstrate that our proposed MHR model achieves state-of-the-art performance, indicating the benefits of hyperbolic representation.**

Received 25 October 2023; revised 16 November 2024; accepted 7 January 2025. Date of publication 14 January 2025; date of current version 7 March 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFC3300200, and in part by the Joint Funds of the National Natural Science Foundation of China under Grant U2336211. It was partly supported in part by the National Natural Science Foundation of China under Grant 62202124. It was also supported in part by the National Research Foundation, in part by Singapore and Infocomm Media Development Authority under its Trust Tech Funding Initiative, in part by the Center for Frontier Artificial Intelligence Research Center, in part by Institute of High Performance Computing, A*STAR, and in part by the College of Computing and Data Science at Nanyang Technological University. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not reflect the views of the National Natural Science Foundation of China, National Research Foundation, Singapore, and Infocomm Media Development Authority. Recommended for acceptance by G. Wang. *(Shanshan Feng and Guoxin Yu are contributed equally to this work.) (Corresponding authors: Han Hu; Hui Lin.)*

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Digital Object Identifier 10.1109/TKDE.2025.3528951

***Index Terms*—Fake news detection, hyperbolic representation, multi-modal learning.**

I. Introduction

FAKE news has emerged as a major challenge in contemporary society, impacting public health, political stability, and social harmony. The rapid growth of the internet and social media platforms, including Twitter and Weibo, has led to an exponential rise in the speed and scale of fake news dissemination. Therefore, detecting fake news on social networks has become an urgent task. Technological advancements have spurred the development of numerous fake news detection methods [1], [2], [3], [4], [5], [6], [7], [8]. Nonetheless, detecting fake news remains a complex challenge. First, fake news often uses techniques like exaggeration, emotional language, and clickbait titles to attract the attention of audiences, leading them to believe or share the content without verifying it thoroughly. Hence, fake news exhibits many similarities in terms of wording and writing style. Second, fake news frequently masquerade as genuine news by incorporating credible media outlets' names, logos, and URLs, or citing fictitious experts, data, or evidence to augment its credibility. As a result, there exist certain correlations between fake and real news. Furthermore, fake news employs diverse media formats, such as texts and images, to enhance its persuasiveness and realism. However, these media contents may have undergone manipulation, splicing, or forgery, deviating from the original sources or facts. These increasingly complex challenges significantly hinder the accurate identification of fake news.

In recent years, numerous studies have been conducted on the detection of fake news. Some early works utilize conventional machine learning approaches [9] to scrutinize rumors' salient attributes in online social media. However, feature engineering is challenging and cannot effectively address the semantic gap issue. More recent works utilize deep learning techniques [10], such as Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN), to automatically extract features, and often achieve superior performance compared to the traditional machine learning approaches. Inspired by the success of Transformers, attention mechanisms have been introduced to enhance fake news detection. For example, Jin et al. [11] propose a multi-modal fusion method through attention mechanisms based on recurrent neural networks to detect fake news. Furthermore, Graph Neural Network (GNN) has been introduced to detect fake news [12], [13], [14], since graphs are suitable for representing

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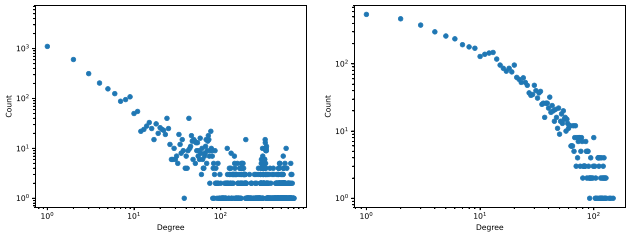

(a) Weibo (b) Pheme

Fig. 1. Distributions of node degrees on Weibo and Pheme. The x-axis presents the node degrees and the y-axis denotes the count of such nodes. The power-law distribution indicates the underlying hierarchical structure.

TABLE I
THE STATISTICS OF DIFFERENT EDGE TYPES WITH VARYING θ VALUES

Dataset	Method	Threshold θ 0.1	0.3	0.5	0.7
Weibo	# R-R	9,075	1,271	735	563
	# F-R	4,999	189	16	7
	# F-F	235,615	107,534	46,565	15,713
Twitter	# R-R	717,642	345,100	200,925	38,474
	# F-R	583,593	2,258	45	5
	# F-F	2,500,540	615,352	138,499	15,513
Pheme	# R-R	168,179	4,071	841	306
	# F-R	142,151	1,705	104	23
	# F-F	108,235	5,504	593	115

TABLE II
δ-HYPERBOLICITY OF THREE GRAPHS

	Weibo	Twitter	Pheme
δ_{avg}	0.154	0.289	0.309
d_{avg}	34.132	159.985	15.301
δ_G	0.009	0.004	0.040

Given a graph, the lower the δ-hyperbolicity, the more tree-like it is.

intricate relationships among multiple entities.The aforementioned methods can be generally categorized into two groups: models based on single-modal information [15], [16], [17], [18] and those based on multi-modal information [1], [2], [19], [20], [21], [22], [23], [24], [25]. Single-modal models predominantly concentrate on the textual or visual features found in the news' headline and body. Conversely, multi-modal methods encompass a wide range of information, such as text and images, to analyze and classify news.

Although existing methods have obtained excellent performance for fake news detection, they treat news as isolated entities and ignore their correlations. To this end, we analyze three commonly used real-world datasets (Weibo [11], Twitter [11], [20] and Pheme [26]) in the fake news detection field and construct graphs to reflect the correlations between news. Note that in the constructed graphs, nodes represent pieces of news and edges denote the calculated similarities between news' content,[1] which is distinct from the context-based graph structures, e.g., user connections, post-user interactions, and propagation structures, in previous studies [4], [12], [13], [14], [27], [28], [29].

The node degree distributions of constructed graphs are shown in Fig. 1 (presented in Section III). From the results, we observe that the node degrees follow a power-law distribution, which indicates that most nodes have a few connections, while a few nodes have a large number of connections. Such imbalanced distribution has not been explored in previous fake news detection studies. Some existing works have shown a high correlation between the power-law distribution of node degree and the underlying *hierarchical structure* [30], [31], [32]. In addition, as reported in Table II, the quantitative analysis of $\delta-$hyperbolicity scores on the built graphs (presented in Section III) also demonstrates that implicit hierarchical structures widely exist in these graphs. However, most existing studies on fake news detection are conducted in the Euclidean space, and cannot capture the underlying complex relationships in the data, particularly the hierarchical structure. To remedy this drawback, we explore detecting fake news in the hyperbolic space, which is more appropriate for handling hierarchical data than in the Euclidean space. Specifically, the hyperbolic space exhibits exponential volume growth [30], [33], while the Euclidean space only exhibits polynomial volume growth, enabling hyperbolic space to utilize a lower dimension to obtain a more comprehensive and efficient information representation.

The data observation and analysis on real-world datasets motivate us to develop a novel Multi-modal Hyperbolic Representation framework (MHR) for detecting fake news. The framework mainly consists of the following components: (1) Graph construction, where each node represents a post and the similarities between nodes are measured according to their TF-IDF scores.[2] By constructing a graph for these nodes, we can capture the correlations between different news. (2) Multi-modal feature extraction, where textual and visual features for each post are extracted using pre-trained XLNet [34] and ViT [35] models, respectively. (3) Lorentzian multi-modal fusion, which integrates the extracted features as the fused node features. Since news usually contains information in terms of multiple modalities, we fully utilize multi-modal information in the process of fake news detection to achieve better detection performance; (4) Fully hyperbolic graph representation learning, where the constructed graph is represented in a hyperbolic space to obtain the node representations. In this way, we can explicitly model the underlying hierarchical structures in the correlations among different news; (5) Fake news detection, where nodes are classified using a fake news detector. This hyperbolic framework is simple yet effective without introducing complicated modules, mainly employing the correlations among news' multi-modal content and the geometrical properties of non-Euclidean spaces. Experiments on three commonly used datasets demonstrate the superiority of our MHR over existing state-of-the-art approaches.

Our main contributions are summarized as follows:

[1]In this work, we calculate the news' similarity by their textual information. We explore various techniques for textual relevance, and the constructed graphs exhibit similar patterns.

[2]The TF-IDF method has a low computational cost and is capable of capturing statistical properties that more sophisticated feature extraction approaches may overlook.

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- We construct graphs for three real-world datasets and analyze their characteristics, showing evidence of underlying hierarchical structures on constructed graphs of the news posts, which has not been investigated in the domain of fake news detection.
- We introduce a new framework, MHR, for fake news detection that represents the relationships between news using a graph while considering the multi-modal features of news by a Lorentzian multi-modal fusion approach.
- We utilize a representation learning method in the hyperbolic space to jointly capture graph structures among news and multi-modal features of news, which is different from existing Euclidean-based methods. To the best of our knowledge, this is the first work to explore graph learning in hyperbolic space for multi-modal fake news detection.
- Extensive empirical comparisons with a variety of counterparts on three widely used datasets demonstrate the superiority of the proposed MHR framework in identifying fake news. Our source code is released at https://github.com/guoxinyu0617/MHR.

II. Related Work

A. Fake News Detection

Detecting fake news is a critical issue that has been explored by various methods and techniques. Initially, researchers rely on manually formulated rules based on common fake news characteristics such as sensational headlines and misleading information [36], [37]. However, these methods are constrained in scalability and require expert knowledge. More recently, a significant increase in research has leveraged deep learning techniques for detecting fake news. These methods can be classified into two categories: *single-modal* and *multi-modal* fake news detection.

Single-modal fake news detection classifies news using only one modality, such as text or images. Over the years, various methods [15], [17], [38], [39], [40] have been proposed based on textual or visual information to detect fake news. For example, Liang et al. [17] approaches the task of text classification by representing an entire corpus as a heterogeneous graph and learning joint word and document embeddings with GCN.

Multi-modal fake news detection employs information from multiple modalities to achieve superior performance [1], [2], [7], [8], [20], [21], [22], [23], [24], [41], [42], [43], [44]. For instance, Qian et al. [1] propose a hierarchical multi-modal contextual attention network that models multilevel semantic relationships and utilizes information from text and images to enhance the accuracy and robustness of fake news detection. Chen et al. [2] suggest a multi-modal approach to detect fake news that accounts for semantic ambiguity between multi-modal content and uses the cross-modal fusion module to capture the correlations between different modalities. Very recently, Wang et al. [43] design a multi-modal masked transformer network for fake news detection by capturing the inter-modality and intra-modality relationships of multi-modal information, and Wu et al. [44] develop a novel cognition-based consistency inference network to model both consistent and inconsistent semantics for multi-modal fake news detection.

In recent years, several studies [3], [4], [6], [12], [13], [14], [27], [28], [29], [45], [46], [47] have investigated the use of graph neural networks (GNNs) for detecting fake news. For instance, Zhang et al. [4] propose SAGNN, an approach based on a simplified GNN method that calculates the degree of interaction between Twitter users and other users for detecting rumors. Moreover, Bian et al. [27] develop a Bi-GCN model that utilizes top-down and bottom-up propagation operations to identify vital rumor features, specifically dispersion and propagation. Most GNN-based methods for fake news detection are context-based, leveraging social context information such as user friendships or tweet-forwarding relationships as a graph. However, content-based approaches [48] have proven more effective in detecting fake news since users and propagation structures become available only after news has spread to the masses [49]. As a content-based method, Qian et al. [12] develop a KMAGCN framework by modeling each post as a graph and using knowledge-aware GCNs to learn representations, where nodes denote words and edges indicate dependency between words in the news post, which is different from our work. In this paper, we propose a novel content-based approach where we construct graphs connecting related posts to leverage their relationships. We generate fused multi-modal features by considering both text and image features of news as node features of the graph. In addition, we explore the hyperbolic space to address the representation learning for graphs, which is different from existing GNN-based methods that commonly use Euclidean spaces.

B. Hyperbolic Representation Learning

Recently, hyperbolic learning techniques have been developed to model complex topological structures of graphs [30], [33], [50], [51], [52], especially the hierarchical structures that commonly exist in information networks. For instance, Zhang et al. [53] develop the HGAT model, which handles highly non-Euclidean structures by computing similarity and attention coefficients between nodes in hyperbolic space. Recently, instead of using the operations in tangent spaces [50], [51], Chen et al. [52] propose a fully hyperbolic framework to build hyperbolic networks based on the Lorentz model by adapting the Lorentz transformation, which can make full use of the hyperbolic spaces to achieve fast and stable performance.

Researchers have started to explore the applicability of learning GNNs in hyperbolic space to achieve better performance on different tasks [54], such as knowledge graphs [55], recommendation sytems [56], [57], and information diffusion [58]. In addition, the hyperbolic representation learning has also been investigated in the word embedding [59], [60], [61], [62], [63] and image embeddeing [64], [65], [66], demonstrating the effectiveness of hyperbolic spaces for multi-media information.

Recently, hyperbolic techniques have been exploited for social media classification tasks. Grover et al. [67] develop a discourse-aware hyperbolic Fourier Co-Attention network for social-text classification. However, their method relies solely

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on single-modal features, and their graph is constructed from public discourses, which differs from our setting. Additionally, they utilize the HGCN [50] in Tangent space to learn hyperbolic node representations, whereas we employ a fully hyperbolic representation model. Nadeem et al. [68] propose a stylometric and semantic similarity-oriented method for multimodal fake news detection. They use the hyperbolic hierarchical attention network [62] to extract stylometric textual features. However, their main detection process operates in Euclidean space and does not consider the correlations among news articles, which is different from our work.

Although hyperbolic graph neural networks (HGNN) have shown remarkable performance in graph analysis, to the best of our knowledge, there is no existing HGNN study for multi-modal fake news detection. In addition, existing HGNN research works usually target at modeling interactions between nodes or capturing a single type of features, and do not consider the complex multi-modal features of nodes, such as textual information and visual information, that are important for recognizing fake news effectively.

III. Data Analysis and Problem Statement

In this section, we process and analyze three datasets commonly used in fake news detection. Based on the results of our observations, we discover that the graphs constructed in this paper may exhibit potential hierarchical structures.

A. Data Analysis

1) Construction of the Graph: In this subsection, we introduce the process of constructing graphs based on real-world datasets (Weibo, Twitter, and Pheme). First, we preprocess the text in the datasets, including tokenization, stop word removal, and punctuation removal. Next, we apply the TF-IDF algorithm[3] to calculate the TF-IDF vector for each news article. TF-IDF is a text feature extraction technique that evaluates the importance of a word in a collection of documents, which provides better interpretability, faster calculation speed, and lower data requirements. We then use the cosine similarity to compute the similarity score between each pair of news articles. A higher similarity score indicates a stronger correlation between two news articles, and an edge is created to connect nodes with strong correlations. Specifically, we utilize a threshold value θ to build the graph: if the similarity between two nodes exceeds θ, an edge is created; otherwise, no edge is formed. The impact of θ will be discussed in the experiments. Note that in our proposed solution, graph construction is a preliminary step before employing the proposed MHR method. We calculate edge weights using cosine similarity in Euclidean space, rather than hyperbolic space, to reflect similarities between news articles, which is more efficient and practical.

2) The Statistics of Constructed Graphs: We examine the statistics of different edge types in the constructed graphs, including R-R, F-R, and F-F. Here, F denotes a false news article, and R indicates a real news article. As illustrated in Table I, we can see that the number of edges decreases with the increase of θ. This is because more edges are filtered out when the threshold is high. Moreover, we observe that the relationships between real news articles and fake news, i.e., F-R, are less frequent than the real-real (R-R) and fake-fake (F-F) relationships. This finding indicates the semantic gaps between false news nodes and real news nodes, which suggests that the neighborhood of news nodes would be beneficial for node classification.

Intuitively, correlations between fake news nodes and real news nodes can be leveraged to differentiate them. Generally, connected neighbors are more likely to exhibit similar behaviors. If a news article has extensive connections to fake news articles, it is more likely to be fake; otherwise, it is likely to be real. Furthermore, fake news and real news nodes share weak correlations, leading to distinct representations through graph learning techniques. This separation of fake and real news representations can effectively support node classification. Hence, incorporating neighborhood information from the constructed graph correlations would aid in distinguishing between real and fake news.

3) Power-Law Distributions of Node Degrees: Next, we analyze the node degree distribution of the graphs constructed from these datasets. Due to space limitations, we only present the distribution of the Weibo and Pheme datasets in Fig. 1.

The distribution of node degree in our analysis follows a power-law distribution, which indicates that most nodes have a few connections, while a few nodes have a large number of connections. Previous research has shown that the power-law distribution of node degree is highly related to the underlying hierarchical structure [30], [31], [32]. In other words, these results suggest the hidden hierarchical structures among these news posts, which have never been studied in the research domain of fake news detection. The power-law distributions of node degrees cannot be adequately captured by Euclidean space due to its polynomial expansion capacity [54], motivating us to explore novel techniques.

4) δ-Hyperbolicity: To quantitatively analyze the hierarchical structure in the Weibo, Twitter, and Pheme graphs, we use δ-hyperbolicity [70]. Following [59], [70], we randomly sample nodes to obtain different quadruples and calculate the average hyperbolicity δ_{avg}. Additionally, we calculate the ratio $\delta_G = 2 \cdot \delta_{avg}/d_{avg}$, which is invariant to metric scaling [59], where d_{avg} represents the average degree between any two nodes in the graph. δ_G can measure how closely a graph resembles a hyperbolic space, with lower δ_G indicating a more hyperbolic graph, i.e., more tree-like data. For completely tree-shaped data, $\delta_G = 0$. We computed the δ_G for the Weibo, Twitter, and Pheme graphs by randomly sampling 50,000 quadruples in the respective graph. The resulting values for δ_{avg}, d_{avg}, and δ_G are shown in Table II.

From the data presented in the Table II, it can be observed that all three graphs exhibit relatively low δ_G values. This result supports that all three graphs may possess latent hierarchical structures [70], as their δ_G values are close to 0. Note that there is no explicit hierarchical structure, such as predefined

[3]we also experimented with other graph construction techniques, such as direct usage of text features obtained from pre-trained models [34], [69] to form the graph, and we find similar characters.

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or extracted tree structures. Previous studies [30], [33], [50] have shown that hyperbolic space is well-suited for modeling implicit hierarchical structures in graphs, leading to improved performance on downstream tasks, such as node classification. This inspires us to explore hyperbolic graph learning for fake news detection, which has not been well studied.

B. Problem Statement

Both qualitative and quantitative analyses of real-world datasets confirm our belief that the news post graph may contain a potential hierarchical structure. Based on this finding, we attempt to investigate how to utilize the correlations among news posts and capture the underlying hierarchical structures for the fake news detection task. Inspired by the success of hyperbolic representation learning [33] for potentially tree-like information networks, we propose to address the fake news detection problem from a new perspective, i.e., using the hyperbolic space to analyze and identify fake news. Notably, this is distinct from existing methods based on Euclidean spaces, and avoids the limitation of the representation ability of Euclidean geometry.

We define fake news detection as a node classification problem in hyperbolic space based on the graph. Given a graph $G = (V, E)$, where V is the set of multi-modal news nodes, and E is the set of edges between multi-modal news nodes. For each news node $v_i \in V$, we extract its textual and visual features and fuse them into a feature vector x_i that represents the news features. The proposed model takes G as input, maps the node features to hyperbolic space, learns the graph representation in hyperbolic space, and then generates a label $Y = 0, 1$ for each node. Here, the label 0 denotes fake news, and 1 represents real news.

IV. Method

In this section, we first introduce some necessary background knowledge about hyperbolic geometry. Then, we present the proposed framework in this paper.

A. Hyperbolic Geometry

Hyperbolic space is a type of Riemannian manifold in geometry with negative constant curvature, which can be described by several models. In this study, we choose the Lorentz model because it has an efficient space for Riemannian optimization and is more numerically stable compared with other models [33]. Due to the length limitations, we only present the relevant concepts of the Lorentz model necessary in this paper. For a more detailed understanding of hyperbolic space, readers can refer to [33], [54].

The Lorentz model of an n-dimensional hyperbolic space is a manifold embedded in the $(n+1)$-dimensional Minkowski space. The n-dimensional Lorentz model $\mathbb{L}_c^n$ with negative curvature c $(c < 0)$ is defined as a Riemannian manifold $(\mathbb{L}_c^n, g_x^L)$, where its metric tensor g_x^L is given by $g_x^L = \text{diag}([-1, 1, \ldots, 1])$ and $\mathbb{L}_c^n = \{\mathbf{x} \in \mathbb{R}^{n+1} :< x, x >_{\mathbb{L}} = \frac{1}{c}\}$. Here, $\mathbf{x} = \{x_0, x_1, x_2, \ldots, x_n\}$ is an n-dimensional vector in the Lorentz model and $< \cdot, \cdot >_{\mathbb{L}}$ denotes the Lorentzian scalar product, which is computed as follows:

$$< \mathbf{x}, \mathbf{y} >_{\mathbb{L}} = \mathbf{x}^T g_x^L \mathbf{y} = -x_0 y_0 + \sum_{i=1}^{n} x_i y_i, \mathbf{x}, \mathbf{y} \in \mathbb{R}^{n+1}. \quad (1)$$

For any $\mathbf{x} \in \mathbb{L}_c^n$, we can get that $x_0 = \sqrt{-\frac{1}{c} + \sum_{i=1}^{n} x_i^2}$.

Given a point $\mathbf{x} \in \mathbb{L}_c^n$, the orthogonal space of $\mathbb{L}_c^n$ is known as the tangent space, which is formally defined as

$$\mathcal{T}_x \mathbb{L}_c^n = \mathbf{y} \in \mathbb{R}^{n+1} | < \mathbf{x}, \mathbf{y} >_{\mathbb{L}} = 0. \quad (2)$$

Note that $\mathcal{T}_x \mathbb{L}_c^n$ is a subspace of Euclidean space $\mathbb{R}^{n+1}$. A common technique used in hyperbolic space is the map operation, which establishes the relationship between a hyperbolic space and its corresponding tangent space. The exponential map $exp_{\mathbf{x}}(\mathbf{z})$ is able to transform a tangent vector $\mathbf{z} \in \mathcal{T}_x \mathbb{L}_c^n$ to $\mathbb{L}_c^n$

$$exp_{\mathbf{x}}(\mathbf{z}) = cosh(\theta)\mathbf{x} + sinh(\theta)\frac{\mathbf{z}}{\theta}, \quad (3)$$

where $\theta = \sqrt{-c}||\mathbf{z}||_{\mathbb{L}}$, $||\mathbf{z}||_{\mathbb{L}} = \sqrt{< \mathbf{z}, \mathbf{z} >_{\mathbb{L}}}$. Its inverse operation is the logarithmic map, which maps hyperbolic vectors to the corresponding tangent space.

In the Lorentz model, a widely used function is the squared Lorentzian distance, which can be utilized to compute the weighted aggregation of points. Given two points $\mathbf{x} \in \mathbb{L}_c^n$ and $\mathbf{y} \in \mathbb{L}_c^n$, the squared Lorentzian distance is

$$D_{sql}(\mathbf{x}, \mathbf{y}) = \frac{2}{c} - 2 < \mathbf{x}, \mathbf{y} >_{\mathbb{L}} . \quad (4)$$

B. Overall Framework

This subsection provides an overview of the proposed MHR framework, which leverages hierarchical relationships between news posts and their multi-modal features to improve the accuracy of fake news detection. By leveraging multi-modal features, MHR offers a comprehensive understanding of news, and through hyperbolic space, the model can effectively capture intricate hierarchical structures. The framework's overall architecture, as illustrated in Fig. 2, comprises the following components:

Construction of the Graph. We treat each news as a node in the graph and use the TF-IDF score of each post to capture the relationships between different posts. The details have been discussed in Section III-A.

Multi-Modal Feature Extraction. For each post, we extract textual features using a pre-trained XLNet model [34] and visual features using a pre-trained ViT model [35].

Lorentzian Multi-modal Fusion. To effectively fuse different extracted features, we design a novel Lorentzian multi-modal fusion module that jointly learns multi-modal fused features for each post.

Fully Hyperbolic Graph Representation Learning. Latent hierarchical information is observed in real-world news datasets. Hence, we develop a fully hyperbolic graph representation learning module to capture both hierarchical structures and node features in the constructed graph.

Fake News Detector. The fake news detector takes node representations learned by the hyperbolic graph neural network as

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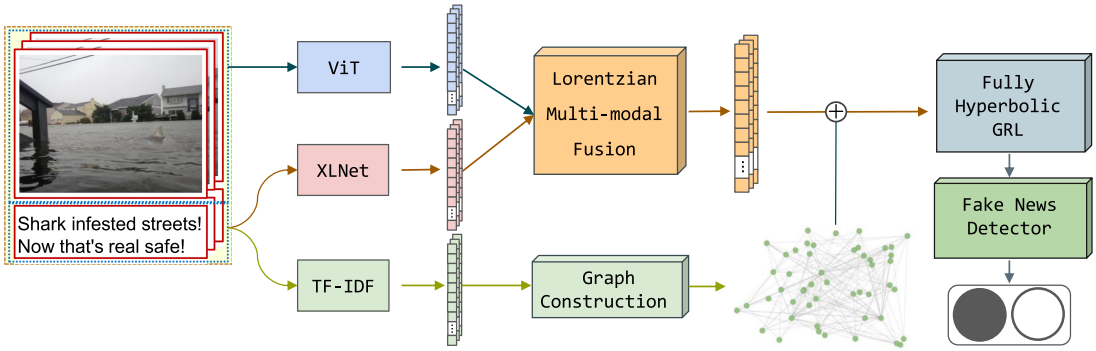

Fig. 2. Overview of the MHR framework. We construct a graph based on the TF-IDF scores of news texts to depict the correlations between them. Textual features are obtained by a pre-trained XLNet model, and visual features are extracted using a pre-trained ViT model. The Lorentzian multi-modal fusion module is utilized to combine these features, which are then input as node features into the fully hyperbolic graph representation learning module for representation learning. Finally, the fake news detector is used to predict whether the given news is true or false.

input and outputs a classification label predicting whether the current news is true or false.

C. Multi-Modal Feature Extraction

1) Textual Feature Extractor: This section presents the extraction of textual features from news articles using a pre-trained XLNet model [34]. Unlike other pre-trained language models such as BERT and GPT, XLNet does not rely on unidirectional or bidirectional context information. It exploits a novel self-regressive modeling method called "Auto-Regressive Factorization" (AR-Factorization) to leverage bidirectional context information without introducing information leakage, leading to better contextual information learning. Moreover, XLNet employs an autoencoder method, enabling the model to better handle incomplete input information.

Let $W = \{w_1, w_2, \ldots, w_m\}$ denote the set of news texts, where w_i represents a news text. The corresponding textual features $\mathbf{T} = \{t_1, t_2, \ldots, t_m\}$ are obtained using the pre-trained XLNet model. Each individual feature t_i corresponds specifically to the text w_i, which is obtained by the XLNet model:

$$\mathbf{T} = \{t_1, t_2, \ldots, t_m\} = XLNet(W). \tag{5}$$

Utilizing the pre-trained XLNet model, we can accurately capture the feature information in news text, leading to improved performance in fake news detection.

2) Visual Feature Extractor: The corresponding images of the news will be input into the visual feature extractor to obtain the image feature V. To extract image features, we utilize a pre-trained ViT model [35]. The ViT model is a cutting-edge computer vision model that uses a transformer-based architecture to process visual data. The fundamental concept behind ViT is to treat an image as a sequence of patches, each of which is considered a token and passed through a transformer-based encoder. This enables the model to capture the spatial relationships between patches, which are vital for image comprehension.

Similarly, we utilize $I = \{i_1, i_2, \ldots, i_m\}$ to represent the set of images corresponding to the news text, where i_i represents an image in the news. We use $\mathbf{V} = \{v_1, v_2, \ldots, v_m\}$ to represent the corresponding visual features, where feature v_i corresponds to image i_i. The visual feature v_i can be obtained through a pre-trained ViT model:

$$\mathbf{V} = \{v_1, v_2, \ldots, v_m\} = ViT(I). \tag{6}$$

The inclusion of visual information as a supplement to textual information permits a comprehensive feature capture of news, resulting in enhancements in detecting fake news.

D. Lorentzian Multi-Modal Fusion

In this section, we present the proposed approach for fusing the extracted textual and visual features in the Lorentz model. We note that previous research has primarily relied on two approaches: concatenating the textual and visual features or employing complex operations such as attention mechanisms to facilitate feature fusion. These conventional methods operate within the Euclidean space, which may not effectively capture the intricate hierarchical structure inherent in the data. However, it is difficult to fuse different modalities into the hyperbolic space due to its geometric properties, i.e., it is infeasible to directly apply existing Euclidean-based techniques in the Lorentz model.

To this end, we design a simple yet effective fusion approach for combining different types of features that are extracted in separate Euclidean spaces. One intuitive idea is directly concatenating features as the fused feature in the Lorentz model. However, this method may have several issues: (1) Some modalities may be more informative than others, and simple concatenation does not consider the relative importance of modalities. (2) Modalities extracted by different pre-trained models may exist in different feature spaces, ultimately leading to unforeseen performance degradation. (3) Direct concatenation of two vectors in a Lorentz model $\mathbb{L}_c^n$ is technically forbidden, since it causes dimension changes and the obtained vector is no longer valid.

Mapping Euclidean Features to the Lorentz Model: To address the aforementioned challenges and elevate the efficacy of feature fusion, we propose a novel approach that transposes the operation to the Lorentz model. More precisely, we first introduce two linear feature mapping matrices in the Euclidean space. Specifically, for different modalities, i.e., text features

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$\mathbf{T}$ and visual features $\mathbf{V}$, we transform these features with two mapping matrices $\mathbf{M}_T$ and $\mathbf{M}_V$, respectively. Then we subject the transformed features to an exponential mapping operation (defined in (3)), assuming the Euclidean extracted features are in the tangent space $\mathcal{T}_0\mathbb{L}$ at the origin point. This transformation results in Lorentzian representations, labeled as $T_{\mathbb{L}}$ for textual features and $V_{\mathbb{L}}$ for visual features,

$$\mathbf{T}_{\mathbb{L}} = exp_0(\mathbf{T} \times \mathbf{M}_T),$$
$$\mathbf{V}_{\mathbb{L}} = exp_0(\mathbf{V} \times \mathbf{M}_V). \tag{7}$$

Here, $exp_0()$ denotes the mapping operation that transfers Euclidean features to the Lorentz model at the origin.

Fuse the Multi-Modal Features: Next, we compute the fusion features within the Lorentz model. Instead of using the Fréchet mean or Einstein midpoint in the Klein model, we calculate the Lorentzian centroid of the mapped textual features $\mathbf{T}_{\mathbb{L}}$ and visual features $\mathbf{V}_{\mathbb{L}}$, which is fast and stable [71]. Specifically, given a news post, we get the centroid of its Lorentzian textual and visual feature vector as the fused features $\mathbf{u}_{fuse}$:

$$\mathbf{u}_{fused} = Centroid\left(\{\mathbf{t}, \mathbf{v}\}, \{0.5, 0.5\}\right). \tag{8}$$

Here, $\mathbf{t}$ and $\mathbf{v}$ is the corresponding Lorentzian vector of the given news post from the $\mathbf{T}_{\mathbb{L}}$ and $\mathbf{V}_{\mathbb{L}}$, respectively.[4] $Centroid()$ denotes the operation of taking the centroid using the squared Lorentzian distance, and $\{0.5, 0.5\}$ means the two vectors are equally weighted.

More generally, given a set of points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ and weights$\{w_1, w_2, \ldots, w_k\}$, the Lorentzian centroid [71] is computed as

$$Centroid\left(\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}, \{w_1, w_2, \ldots, w_k\}\right)$$
$$= \frac{\sum_{i=1}^{k} w_i \mathbf{x}_i}{\sqrt{-c} \mid \| \sum_{i=1}^{k} w_i \mathbf{x}_i \|_{\mathbb{L}} \mid}. \tag{9}$$

We can learn that (8) is a simple case of the above equation. In this manner, we fuse both the textual features and visual features into a unified Lorentzian feature vector.

Linear Transformation: To map the fused Lorentzian vector to a low-dimensional Lorentzian space, we add a Lorentzian linear layer $f_{\mathbb{L}}^{Linear}$ to derive the final fused features $\mathbf{x}_u^0 = f_{\mathbb{L}}^{Linear}(\mathbf{u}_{fused})$. Instead of the conventional hyperbolic linear transformations in tangent space [50], we utilize the Lorentzian linear transformations [52].

Specifically, given an vector $\mathbf{x} \in \mathbb{L}_c^n$, we attempt to learn a transformation matrix $\mathbf{M} \in \mathbb{R}^{(n+1)\times(d+1)}$, where $\mathbf{M}$ consists of two components: $\mathbf{v} \in \mathbb{R}^{d+1}$ and $\mathbf{W} \in \mathbb{R}^{n\times(d+1)}$. d denotes the number of dimensions for our target hyperbolic space. Hence, we can get the linearly transformed vector $f'_{\mathbb{L}}(\mathbf{M})\mathbf{x} \in \mathbb{L}_c^d$, where $f'_{\mathbb{L}}(\mathbf{M})$ is defined as

$$f'_{\mathbb{L}}(\mathbf{M}) = f'_{\mathbb{L}}\left(\begin{bmatrix}\mathbf{v}^T \\ \mathbf{W}\end{bmatrix}\right) = \begin{bmatrix}\frac{\sqrt{\|\mathbf{W}\mathbf{x}\|^2 - 1/c}}{\mathbf{v}^T\mathbf{x}}\mathbf{v}^T \\ \mathbf{W}\end{bmatrix}. \tag{10}$$

[4]In our experiments, the outputs of ViT and XLNet are both 768-dimensional features. Hence, the fused vector is a 768-dimensional Lorentzian vector.

In this manner, we obtain the final fused vector $\mathbf{x}^0 = f'_{\mathbb{L}}(\mathbf{M})\mathbf{u}_{fused}$, which will be used as the input node features in the next graph representation learning module. Integrating multi-modal features enables us to provide a more comprehensive and accurate description of event authenticity, enhancing the detecting performance of fake news.

E. Fully Hyperbolic Graph Representation Learning

To enhance node representation and improve model accuracy and computational efficiency, we utilize a fully hyperbolic approach for learning node representations, which captures hierarchical structures among news articles and their fused feature vectors.

Graph neural networks are widely used for learning graph representations in the Euclidean space, which updates the feature representation of each node through the following steps:

$$h_i^l = W^l x_i^{l-1} + b^l, \;\; x_i^l = \sigma\left(h_i^l + \sum_{j\in\mathcal{N}_i} w_{i,j} h_j^l\right). \tag{11}$$

W^l and b^l are the weight and bias parameters of the lth layer, respectively. x_i^{l-1} denotes the representation of node i in the $(l-1)$th layer, $\mathcal{N}_i$ denotes the set of neighboring nodes of node i, and $\sigma(\cdot)$ represents a non-linear activation function. The aggregation weight $w_{i,j}$ can be computed in various ways [72], [73].

However, the operational rules in hyperbolic space differ from those in the Euclidean space, requiring the redefinition of these operations. In our framework, instead of employing the hyperbolic graph neural network [50] that mainly uses operations in the tangent space, which is not completely hyperbolic and leads to performance loss, we adopt fully hyperbolic operations for modeling the graph of news.

Feature Transform: We first present the general formula of linear layers based on Equation 10, which incorporates transformation, activation, and normalization,

$$\mathbf{y} = HL(\mathbf{x}) = \begin{bmatrix}\sqrt{\|\psi(\mathbf{W}\mathbf{x}, \mathbf{v})\|^2 - 1/c} \\ \psi(\mathbf{W}\mathbf{x}, \mathbf{v})\end{bmatrix}, \tag{12}$$

where $\mathbf{x} \in \mathbb{L}_c^d$, $\mathbf{v} \in \mathbb{R}^{d+1}$, $\mathbf{W} \in \mathbb{R}^{m\times(d+1)}$. The $\psi(\cdot)$ denotes an operation function of activation and normalization, defined as

$$\psi(\mathbf{W}\mathbf{x}, \mathbf{v}) = \frac{\gamma\sigma(\mathbf{v}^T\mathbf{x} + b')}{\|\mathbf{W}h(\mathbf{x}) + b\|}\left(\mathbf{W}h(\mathbf{x}) + b\right), \tag{13}$$

where $\sigma(\cdot)$ indicates the sigmoid function, both b and b' denote bias terms, $\gamma > 0$ determines the scaling parameter, and $h()$ is the activation function.

Neighborhood Aggregation: Aggregation is a crucial step in GNN, which determines how information is propagated in the graph. In this work, we exploit an attention-based aggregation mechanism to capture neighborhood structures and features. The key challenge is to learn the neighbors' importance and then calculate the weighted mean in the Lorentz model, as the conventional attention approach cannot be directly applied.

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Given a query set $\mathcal{Q} = \{\mathbf{q}_1, \ldots, \mathbf{q}_{|\mathcal{Q}|}\}$, key set $\mathcal{K} = \{\mathbf{k}_1, \ldots, \mathbf{k}_{|\mathcal{K}|}\}$, and value set $\mathcal{V} = \{\mathbf{v}_1, \ldots, \mathbf{v}_{|\mathcal{V}|}\}$, we utilize the squared Lorentzain distance (4) between $\mathbf{q}_i$ and $\mathbf{k}_j$ to calculate the weight w_{ij}:

$$w_{ij} = \frac{exp\left(-D_{sql}(\mathbf{q}_i, \mathbf{k}_j)/\sqrt{n}\right)}{\sum_{k=1}^{|\mathcal{K}|} exp\left(-D_{sql}(\mathbf{q}_i, \mathbf{k}_k)/\sqrt{n}\right)}. \tag{14}$$

Here, n is the number of dimensions for the points. Then, we use the Centroid (9) to obtain the aggregated point:

$$\mathbf{u}_i = \frac{\sum_{j=1}^{|\mathcal{K}|} w_{ij}\mathbf{v}_j}{\sqrt{-c}\,|\,\|\sum_{j=1}^{|\mathcal{K}|} w_{ij}\mathbf{v}_j\|_{\mathbb{L}}\,|}. \tag{15}$$

The above mechanism can be denoted as

$$\{\mathbf{u}_1, \mathbf{u}_2, \ldots\} = ATT\left(HL_{\mathcal{Q}}(\mathcal{Q}), HL_{\mathcal{K}}(\mathcal{K}), HL_{\mathcal{V}}(\mathcal{V})\right), \tag{16}$$

where $HL_{\mathcal{Q}}, HL_{\mathcal{K}}, HL_{\mathcal{V}}$ indicate the corresponding hyperbolic linear layer for query, key, and value, respectively.

By considering multiple layers, the overall architecture of graph representation learning can be formulated as

$$\mathbf{X}_i^l = ATT\left(HL_{\mathcal{Q}}(\mathbf{X}_i^{l-1}), \{HL_{\mathcal{K}}(\mathbf{X}_{j\in\mathcal{N}_i}^{l-1})\}, \{HL_{\mathcal{V}}(\mathbf{X}_{j\in\mathcal{N}_i}^{l-1})\}\right), \tag{17}$$

where $\mathbf{X}_i^l$ is the representation of i-node at the layer l, and $\mathcal{N}_i$ refers to neighbors of node i. In this manner, we learn the news posts' Lorentzian representations $\mathbf{X}$, which can be used for identifying fake news.

Overall, our work builds upon the hyperbolic neural networks proposed by [52]. However, it differs from [52] in two key aspects. First, they primarily address different tasks. Second, we additionally employ the Lorentzian multi-modal fusion module to integrate various node features.

F. Fake News Detector

In this subsection, we present the fake news detector, which receives the learned Lorentzian representations $\mathbf{X}_i^l$ as input and predicts whether the current news is true or false. This task can be regarded as a classification task. Note that we cannot straightforwardly apply the fully connected layer to predict labels as commonly used in existing Euclidean fake news detection methods [1], [44], due to the hyperbolic geometric limitations. Hence, we assign a learnable representation $\mathbf{G}_{C_k}$ for each class C_k. Then we can calculate the squared Lorentzian distance between $\mathbf{G}_{C_k}$ and node representation $\mathbf{X}_i^l$. The probability of the ith news belonging to class C_k is computed by the Fermi-Dirac decoder [30]

$$P_{i,C_k} = \frac{1}{1 + exp\left((D_{sql}(\mathbf{X}_i^l, \mathbf{G}_{C_k}) - r)/t\right)}, \tag{18}$$

where r and t are hyperparameters.

Furthermore, let $\hat{P}_i$ represent the predicted probability of the ith news and let Y_i denote the true label of the ith news. Then we can calculate the detection loss by employing the cross-entropy:

$$\mathcal{L}(\Theta) = \sum_{i=1}^{m} -[Y_i log(\hat{P}_i) + (1 - Y_i) log(1 - \hat{P}_i)], \tag{19}$$

where m is the number of news in the dataset.

TABLE III
STATISTICS OF THE EMPLOYED REAL-WORLD DATASETS

News	Weibo	Twitter	Pheme
# of Fake News	4,749	7,898	1,972
# of Real News	4,779	6,026	3,830
# of Images	9,528	514	3,670

V. EXPERIMENTS

A. Experimental Setup

1) Datasets: In this paper, we utilize three datasets commonly used in the field of fake news identification: Weibo [11], Twitter [11], [20] and Pheme [26]. The Weibo dataset comprises all the fake news verified by Weibo's official debunking system and the real news verified by China's authoritative news agency, Xinhua News Agency.[5] The Twitter dataset is sourced from the MediaEval Verifying Multimedia Use benchmark, which provides text information and visual information for tweets on Twitter. The Pheme dataset is collected based on 5 breaking news, each containing a set of posts. Each dataset includes a large number of texts and images with labels. Following the [1], [19], we partition the Weibo and Pheme datasets into training, validation, and test sets in a ratio of 7:1:2. The Twitter dataset is composed of both a development and test set, and we employ the development set as the training set and the test set for testing. Statistical information for three datasets is presented in Table III.

2) Evaluation Metrics: In this paper, we employ four metrics to assess the performance of our model: accuracy, precision, recall, and F1 score. Accuracy is a common evaluation metric that can measure the correctness of the model's predictions across all samples, it can be influenced by imbalanced category distributions in binary classification problems such as fake news detection. Hence, to comprehensively evaluate the model's performance, we also considere precision and recall, as well as the F1 score, which is a composite metric of these two measures.

3) Implementation Details: To obtain the multi-modal representation of news in the feature fusion stage, we leverage pre-trained models: the XLNet model [34] to extract text features and the ViT model [35] to extract image features. Both text and image features are set to a dimension of 768. For news articles without images, we utilize a zero-vector as the image feature, which is a commonly used method for handling missing values. In the hyperbolic graph representation learning stage, we follow the methodology of [52] and utilize a hyperparameter search method to determine the initial learning rate, dropout, weight decay, number of layers, and activation function. To optimize our model's hyperparameters, we employ Radam as the optimizer.

4) Baselines: To evaluate the effectiveness of the proposed framework in this paper, we compare it against two types of baselines: single-modal models and multi-modal models.

Single-modal Models: Specifically, we first compare the performance of our method with that of four single-modal models that only use one type of information in the news. (1) *SVM-TS* [15], which detects fake news on Weibo by utilizing heuristic rules and a linear SVM classifier. (2) *GRU* [16], which is based

[5][Online]. Available: http://www.xinhuanet.com/

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on RNN and employs multilayer GRU networks to effectively model news as a variable-length time series. (3) *TextGCN* [17], which approaches the text classification task by representing an entire corpus as a heterogeneous graph and learning joint word and document embeddings with GCN. (4) *CNN* [18], which uses CNN to automatically obtain key features of both misinformation and truth information.

Multi-modal Models: We also compare our proposed fake news detection method with nine recent multi-modal models. These models employ multiple sources of information to address the task of fake news detection. (1) *att_RNN* [11], which proposes an innovative RNN with an attention mechanism for effective multi-modal feature fusion. For a fair comparison, this paper removes the information of social context. (2) *EANN* [19], which proposes a multi-modal fake news detection method that relies on event adversarial neural networks. The event discriminator enhances the detection efficiency for new events. Similarly, for fairness of the comparison, this paper does not include the event discriminator. (3) *MVAE* [20], which employs a multi-modal variational autoencoder to extract features from both text and images. (4) *Spotfake* [74], whose primary contribution lies in its use of pre-trained models. Specifically, pre-trained BERT is utilized to extract text features, while pre-trained VGG19 is used to extract image features. (5) *Spotfake+* [75], which is an enhanced version of SpotFake that utilizes an XLNet model to extract textual features. (6) *SAFE* [76], which generates a joint representation of textual and visual features for each article, and measures the similarity using the cosine function. (7) *HMCAN* [1], which proposes a multi-modal contextual attention network that leverages complementary data from different modalities to address the challenge of multi-level semantic relationships in fake news detection. (8) *HCCIN* [44], which utilizes a cross-modal alignment layer to learn consistent semantics between textual and visual information, and develops collaborative inference layers to identify the inconsistent semantics between news and comments. To make a fair comparison, we do not consider the comments information in our experiments. (9) *MMTN* [43], which proposes a masked transformer network to fuse visual and textual features, and mask the irrelevant context between modalities. It adopts a positive unlabeled learning technique to learn the detector for fake news.

For the proposed MHR model, we additionally consider another version, namely *MHR(HGCN)*, which is also implemented in hyperbolic spaces. Unlike the MHR model stated in Section IV, it utilizes the Hyperbolic Graph Convolutional Neural Networks (HGCN) [50] to learn node representations. The extracted textual and visual features are fused in the Euclidean space, and then the fused features are mapped to the Hyperbolic space by the exponential map function [77]. After that, hyperbolic node representations are learned using the tangent space operations [50], including feature transformation, information aggregation, and feature activation. Notably, this version is not a fully hyperbolic approach, since it exploits the operations in the Euclidean tangent space. In this work, we explore the fully hyperbolic learning method via Lorentzian models for fake news detection.

Note that the TextGCN [17] approach is also a GNN-based fake news detection method. It models sentences within a single news article as a graph and does not consider the correlations that exist between different articles, which is dissimilar from this work.

B. Results and Analyses

In this section, we present and analyze the experimental results. Table IV shows the experimental results on the Weibo, Twitter, and Pheme datasets. Based on the results, the following conclusions can be drawn: (1) Among all the evaluated methods, SVM-TS performs the worst, with an accuracy of only around 0.5 on the Twitter dataset. This indicates that hand-crafted features are weak and insufficient for identifying fake news. (2) Deep learning-based single-modality fake news detection models (GRU, TextGCN, CNN) outperform SVM-TS, as they can automatically extract complex features from data using multiple layers of neural networks. (3) Compared with other single-modality deep-learning methods (GRU and CNN), TextGCN is able to achieve much better experimental results. The performance of TextGCN indicates that using a graph as a data structure can capture word co-occurrence and document-word relationships, which is useful for detecting fake news. (4) Generally, multi-modal models perform better than single-modal models, suggesting that using multi-modal information can improve the effectiveness of fake news detection. (5) The MHR(HGCN) can achieve prominent performance on three datasets, surpassing various Euclidean multi-modal models. However, its performance is slightly worse than the proposed MHR method, which indicates the benefits of fully hyperbolic graph representation learning in the Lorentzian model. (6) Overall, the proposed MHR outperforms all baselines on three evaluated datasets. In particular, MHR obtains 4.7% and 4.5% higher accuracy scores than the best scores of baselines on the Weibo and Twitter datasets, respectively. For the key measure metrics (Accuracy, F1 on fake news, and F1 on real news), all reported improvements over baseline methods in this task are statistically significant with p-value < 0.01.

It is worth noting that our proposed MHR does not employ sophisticated alignment modules as baselines [1], [43], [44], and it merely utilizes the simple linear feature alignment. The key technical difference is using the hyperbolic representation based on the constructed news graph, which can effectively capture node features and complex graph structures. To sum up, the proposed MHR model and its variant MHR(HGCN) are able to achieve superior performance over the existing Euclidean detection approaches on the three datasets. Empirical results validate that multi-modal information can effectively assist the model in detecting fake news, while hyperbolic space can better represent and learn graph data of news articles.

C. Ablation Study

We analyze the results of ablation experiments conducted to evaluate the effectiveness of the proposed MHR framework for detecting fake news. The framework mainly incorporates

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TABLE IV
EXPERIMENTAL RESULTS OF MODELS ON THE THREE DATASETS

Dataset	Method	Accuracy	Fake News			Real News		
			Precision	Recall	F1	Precision	Recall	F1
Weibo	SVM-TS	0.640	0.741	0.573	0.646	0.651	0.798	0.711
	GRU	0.702	0.671	0.794	0.727	0.747	0.609	0.671
	TextGCN	0.787	**0.975**	0.573	0.727	0.712	**0.985**	0.827
	CNN	0.740	0.736	0.756	0.744	0.747	0.723	0.735
	att_RNN	0.772	0.854	0.656	0.742	0.720	0.889	0.795
	EANN	0.782	0.827	0.697	0.756	0.752	0.863	0.804
	MVAE	0.824	0.854	0.769	0.809	0.802	0.875	0.837
	SpotFake	0.892	0.902	**0.964**	0.932	0.847	0.656	0.739
	SpotFake+	0.870	0.887	0.849	0.868	0.855	0.892	0.873
	SAFE	0.763	0.833	0.659	0.736	0.717	0.868	0.785
	HMCAN	0.885	0.920	0.845	0.881	0.856	0.926	0.890
	HCCIN	0.857	0.854	0.885	0.869	0.820	0.871	0.845
	MMTN	0.889	0.886	0.893	0.889	0.892	0.885	0.893
	MHR(HGCN)	0.928	0.924	0.932	0.928	**0.933**	0.925	0.929
	MHR	**0.933**	0.951	0.921	**0.936**	0.918	0.949	**0.933**
	(Std Dev)	(0.003)	(0.004)	(0.006)	(0.003)	(0.007)	(0.003)	(0.004)
Twitter	SVM-TS	0.529	0.488	0.497	0.496	0.565	0.556	0.561
	GRU	0.634	0.581	0.812	0.677	0.758	0.505	0.604
	TextGCN	0.703	0.808	0.365	0.503	0.680	0.939	0.779
	CNN	0.549	0.508	0.597	0.549	0.598	0.509	0.550
	SAFE	0.766	0.777	0.795	0.786	0.752	0.731	0.742
	att_RNN	0.664	0.749	0.615	0.676	0.589	0.728	0.651
	EANN	0.648	0.810	0.498	0.617	0.584	0.759	0.660
	MVAE	0.745	0.801	0.719	0.758	0.689	0.777	0.730
	SpotFake	0.777	0.751	0.900	0.820	0.832	0.606	0.701
	SpotFake+	0.790	0.793	0.827	0.810	0.786	0.747	0.766
	HMCAN	0.897	**0.971**	0.801	0.878	0.853	**0.979**	0.912
	HCCIN	0.821	0.834	0.833	0.833	0.773	0.912	0.837
	MMTN	0.903	0.870	0.917	0.892	0.927	0.881	0.903
	MHR(HGCN)	0.932	0.942	0.923	0.932	0.921	0.941	0.931
	MHR	**0.950**	0.973	**0.930**	**0.951**	**0.927**	0.972	**0.949**
	(Std Dev)	(0.005)	(0.008)	(0.007)	(0.006)	(0.008)	(0.008)	(0.006)
Pheme	SVM-TS	0.639	0.546	0.576	0.560	0.729	0.705	0.717
	GRU	0.832	0.782	0.712	0.745	0.855	0.896	0.865
	TextGCN	0.828	0.775	0.735	0.737	0.827	0.828	0.828
	CNN	0.779	0.732	0.606	0.663	0.799	0.875	0.835
	att_RNN	0.850	0.791	0.749	0.770	0.876	0.899	0.888
	EANN	0.681	0.685	0.664	0.694	0.701	0.750	0.747
	MVAE	0.852	0.806	0.719	0.760	0.871	0.917	0.893
	SpotFake	0.823	0.743	0.745	0.744	0.864	0.863	0.863
	SpotFake+	0.800	0.730	0.668	0.697	0.832	0.869	0.850
	SAFE	0.811	0.827	0.559	0.667	0.806	**0.940**	0.866
	HMCAN	0.881	0.830	0.838	0.834	0.910	0.905	0.907
	HCCIN	0.849	0.772	0.769	0.770	0.863	0.870	0.866
	MMTN	0.887	0.870	0.789	0.827	0.895	0.938	**0.915**
	MHR(HGCN)	0.858	0.843	0.869	0.856	0.873	0.847	0.860
	MHR	**0.904**	**0.882**	**0.923**	**0.902**	**0.926**	0.887	0.906
	(Std Dev)	(0.006)	(0.024)	(0.013)	(0.007)	(0.016)	(0.019)	(0.005)

The best results are highlighted in boldface, and the second best results are underlined.

multi-modal feature learning and hyperbolic space graph representation. To demonstrate the impact of different components, we designed several variants of MHR and evaluated their performance in terms of node features, fusion features, graph structure, and hyperbolic space. Six variants of MHR are considered. (1) *MHR* \ *F:* exploits only the graph structure and no textual or visual features. (2) *MHR* \ *T:* uses only visual features as the features of nodes in the graph and no textual features. (3) *MHR* \ *V:* uses only textual features as the features of nodes in the graph and no visual features. (4) *MHR* \ *H:* performs graph representation learning in Euclidean space rather than in hyperbolic space. (5) *MHR* \ *G:* does not employ the graph representation learning, and solely uses textual and visual fusion features to detect fake news. (6) *MHR* \ *LF:* does not exploit the Lorentzian Multi-modal fusion module (Section IV-D). Instead of fusing features in the Lorentzian space, it first fuses the extracted multi-modal features in Euclidean space and then maps them to the Lorentzian model. The results of the ablation experiments are presented in Table V.

Impact of Node Features: We compare the performance of MHR and MHR \ F on three datasets. The experimental results indicate that MHR outperforms MHR \ F, demonstrating the presence of node features can improve the performance of the model by enhancing the graph's representational power.

Impact of Fusion Features: We evaluate the impact of multi-modal features by comparing the performance of MHR, MHR \

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TABLE V
RESULTS OF COMPARISON AMONG DIFFERENT VARIANTS IN MHR ON THE THREE DATASETS

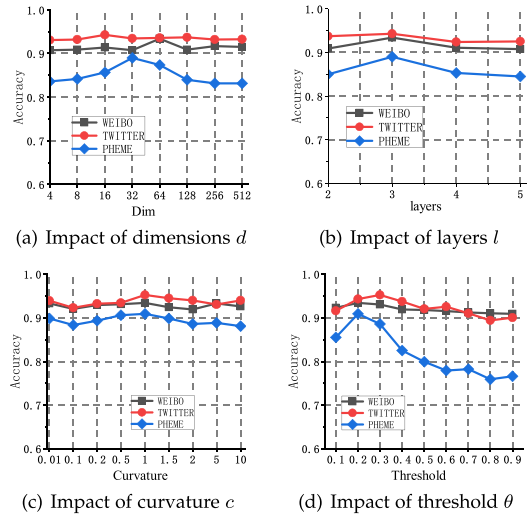
Dataset	Metric	\F	\T	\V	\H	\G	\LF	MHR
Weibo	Acc	0.791	0.892	0.924	0.930	0.929	0.924	**0.933**
	Pre(F)	0.629	0.888	0.909	0.948	0.940	0.944	0.951
	Rec(F)	0.930	0.895	0.936	0.914	0.930	0.907	0.921
	F1(F)	0.751	0.892	0.923	0.931	0.919	0.925	**0.936**
	Pre(R)	0.953	0.896	0.938	0.911	0.919	0.903	0.918
	Rec(R)	0.720	0.889	0.912	0.950	0.939	0.942	0.949
	F1(R)	0.820	0.893	0.925	0.930	0.929	0.922	**0.933**
Twitter	Acc	0.897	0.933	0.894	0.920	0.932	0.924	**0.950**
	Pre(F)	0.899	0.959	0.943	0.935	0.977	0.981	0.973
	Rec(F)	0.895	0.912	0.858	0.908	0.897	0.882	0.930
	F1(F)	0.897	0.935	0.899	0.921	0.935	0.929	**0.951**
	Pre(R)	0.895	0.908	0.844	0.905	0.888	0.868	0.927
	Rec(R)	0.898	0.957	0.937	0.933	0.974	0.978	0.972
	F1(R)	0.897	0.932	0.888	0.919	0.929	0.920	**0.949**
Pheme	Acc	0.867	0.858	0.873	0.865	0.860	0.853	**0.904**
	Pre(F)	0.845	0.807	0.843	0.825	0.853	0.843	0.882
	Rec(F)	0.883	0.898	0.897	0.898	0.866	0.860	0.923
	F1(F)	0.864	0.850	0.869	0.860	0.859	0.851	**0.902**
	Pre(R)	0.888	0.909	0.904	0.906	0.868	0.863	0.926
	Rec(R)	0.852	0.825	0.852	0.838	0.855	0.846	0.887
	F1(R)	0.870	0.865	0.877	0.871	0.861	0.854	**0.906**

Here, \F, \T, \V, \H, \G, and \LF indicate MHR\F, MHR\T, MHR\V, MHR\H, MHR\G and MHR\LF, respectively. (F) denotes the metric for fake news and (R) is for real news

T, and MHR \ V on three datasets. The results show that using multi-modal features enhances the detection performance, as MHR outperforms MHR \ T and MHR \ V in terms of accuracy and F1 scores. Multi-modal features indeed provide more comprehensive information about nodes and reduce the impact of noise and missing data, enhancing the model's robustness.

Impact of Hyperbolic Space: We examine the performance of MHR and MHR \ H to assess the impact of performing graph representation learning in hyperbolic space. Our experimental results show that using hyperbolic space consistently improves the performance of the model, validating our hypothesis that hyperbolic space can effectively characterize both the local connections and global hierarchical structures. Our empirical results align with previous studies [33], [54], [78], [79], showing that hyperbolic space is more expressive than Euclidean space for capturing implicit hierarchical structures. This is because hyperbolic space has exponential expansion capacity due to its unique geometric properties, while Euclidean space only exhibits polynomial expansion.

Impact of Graph Structure: We further evaluate the contribution of using the constructed graph for fake news detection. The MHR \ G model directly exploits fused features of news articles, without considering their connections. As shown in Table V, the performance of MHR \ G is worse than the original MHR model, indicating that the constructed graph helps to identify fake news. This is because, through graph representation learning on the correlations, nodes are more likely to have similar representations to their neighbors. Real news articles tend to have similar representations since they are more likely to be connected to other real news articles as shown in Table I. Similarly, fake news nodes also tend to cluster together and share similar representations. Furthermore, due to the relatively weak correlations between fake news and real news nodes, their learned node representations are distinguishable.



(a) Impact of dimensions d (b) Impact of layers l

(c) Impact of curvature c (d) Impact of threshold θ

Fig. 3. Effects of parameters on three datasets.

Therefore, the separation of fake and real news representations would lead to improved detection performance.

Impact of Lorentzian Fusion: We investigate the benefit of the developed Lorentzian multi-modal fusion module. Comparing MHR \ LF and MHR shows that the Lorentzian fusion approach can effectively integrate different modality features, improving fake news detection performance.

D. Parameter Sensitivity Analyses

Additionally, to investigate the effects of the latent node feature embedding dimension d, the number of layers $layers$, curvature parameter c, and threshold θ in graph construction on the proposed framework's performance, we perform experiments on the three datasets. The results of the parameter impacts are presented in Fig. 3.

Impact of the number of dimensions: We study the parameter sensitivity by varying the d from 4 to 512. The results indicate that MHR's performance can be improved by increasing the number of dimensions, but becomes saturated when d equals 16 or 32, after which the performance stops growing. Another finding is that MHR can achieve remarkable performance with relatively low dimensions, which demonstrates the strong representative power of hyperbolic space for fake news detection. In addition, large embedding dimensions will consume more resources and suffer from lower computational efficiency. Thus, we can set the number of dimensions as 16in practical applications to balance performance and efficiency.

Impact of the number of layers: We examine the performance of MHR by varying its number of layers l from 2 to 5. As shown in Fig. 3(b), we observe that the optimal scores can be achieved when the l equals 3. Exceeding this value can result in negative impacts. The default number of layers can be set as 3 for optimal performance and computational efficiency.

Impact of curvature parameter c: We investigate the effect of curvature parameter c by varying its value from 0.01 to 10. As illustrated in Fig. 3(c), the performance with different c is

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TABLE VI
THE PERFORMANCE OF DIFFERENT METHODS FOR GRAPH CONSTRUCTION ON THE WEIBO DATASET

Method	Acc	Fake News			Real News		
		Pre	Rec	F1	Pre	Rec	F1
(1)Image	0.868	0.972	0.805	0.88	0.764	0.964	0.853
(2)Image+Text	0.878	0.961	0.825	0.888	0.796	0.953	0.867
(3)BOW	0.933	0.954	0.915	0.934	0.912	0.952	0.931
(4)BERT	0.927	0.938	0.919	0.928	0.917	0.937	0.927
(5)XLNet	0.922	0.931	0.915	0.923	0.914	0.929	0.921
(6)TF-IDF	0.933	0.951	0.921	0.936	0.918	0.949	0.933

stable. The optimal results can be obtained when $c = 1.0$, which is also a widely used parameter setting in many hyperbolic graph learning models.

Impact of Threshold θ: Fig. 3(d) shows the performance with different threshold θ values. The performance of MHR first improves as θ increases, then decreases as θ continues to rise. This is because too many connections (low θ) introduce noise, while too few connections (high θ) fail to capture relationships between news articles. The best performance is achieved when θ is 0.2 or 0.3. In our experiments, we set $\theta = 0.3$ for the Weibo and Twitter datasets, and $\theta = 0.2$ for the Pheme dataset.

Impact of Graph Construction Methods: We examine different methods for computing correlations between news articles. As introduced in Section III-A, we use the TF-IDF algorithm as the default method to extract node features. Additionally, we evaluate other methods empirically. Specifically, we compare with five graph construction methods: (1) *Image:* uses only visual features extracted by [35] as node features. (2) *Text+Image:* directly concatenates visual and textual features. (3) *BOW:* uses the bag-of-words model for textual feature extraction. (4) *BERT:* uses BERT [69] for textual feature extraction. (5) *XLNet:* employs the XLNet [34] model to textual features as node features. Note that methods (3-5) use only textual features to calculate node similarity. As reported in Table VI, the performance of methods (1) and (2) is significantly worse than that of the text-only methods. This indicates that visual features are not beneficial for building correlations among news articles, possibly because the semantic gaps in visual features are larger than those in textual features. Additionally, the performance of methods (3-5) is similar to the TF-IDF approach, suggesting that different textual feature extraction methods yield consistent performance. We conducted experiments on two other datasets and obtained similar results. Overall, TF-IDF slightly outperforms other text-only methods. Therefore, we use the TF-IDF method by default in our experiments.

VI. CONCLUSION

Existing fake news detection methods often treat each news as an isolated entity and ignore the correlations between news. Based on the analysis of real-world datasets, we find that there exist hierarchical structures among news that have been ignored by existing works. Distinct from conventional Euclidean methods, this paper introduces a novel Multi-modal Hyperbolic Representation Framework (MHR) for fake news detection. Specifically, we construct a graph based on the correlations between news and consider the multi-modal features of news. We extract the textual and visual information of news and leverage a Lorentzian multi-modal fusion module to obtain the node features in the constructed graph. Moreover, we exploit a fully hyperbolic graph learning module to get node representations for detecting fake news, which is advantageous in handling hierarchical structures and capturing node features. The experimental results on three publicly available datasets demonstrate that our proposed MHR outperforms various recent counterparts. In the future, we intend to leverage information from more modalities and design more effective representation learning algorithms in hyperbolic space to enhance the performance of fake news detection.

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